

Cache-Aware Winograd Implementation

Combined Experimental Results and Platform Metadata Report

1. Hardware & Software Platform Profile

Property	Value
OS / Architecture	Darwin / arm64
CPU Model	arm
Python Version	3.13.2 (v3.13.2:4f8bb3947cf, Feb 4 2025, 11:51:10) [Clang 15.0.0 (clang-1500.3.9.4)]
Logical / Physical Cores	12 / 12
L1 Data Cache Size	64 KB (65536 B)
L2 Cache Size	4096 KB (4194304 B)
Cache Line Size	128 B
Git Commit Reference	4c42e69c
Execution Timestamp	2026-03-22T17:58:51.955551

2. End-to-End Layer Run Status

mode	status	reason
end-to-end	unsupported	missing_dependencies

Note: Full end-to-end framework benchmarks were reported as unsupported due to missing system-level dependencies for PyTorch/ONNX Runtime complete models on this specific environment, shifting the focus to high-fidelity microbenchmarks.

3. Autotiler Decision Configurations

Workload	Target Cache	Selected Tile	Working Set (B)	Reuse Score
C_in=16, C_out=32	32 KB	F(2,3)	18,560	0.4414
C_in=32, C_out=16	32 KB	F(2,3)	35,968	0.1139
C_in=32, C_out=32	32 KB	F(2,3)	35,968	0.2278
C_in=32, C_out=64	32 KB	F(2,3)	35,968	0.4555
C_in=64, C_out=32	32 KB	F(2,3)	70,784	0.1157
C_in=64, C_out=64	32 KB	F(2,3)	70,784	0.2315
C_in=128, C_out=128	32 KB	F(2,3)	140,416	0.2334
C_in=16, C_out=32	64 KB	F(4,3)	41,760	1.5657
C_in=32, C_out=16	64 KB	F(2,3)	35,968	0.2278
C_in=32, C_out=32	64 KB	F(2,3)	35,968	0.4555
C_in=32, C_out=64	64 KB	F(2,3)	35,968	0.911
C_in=64, C_out=32	64 KB	F(2,3)	70,784	0.2315
C_in=64, C_out=64	64 KB	F(2,3)	70,784	0.4629
C_in=128, C_out=128	64 KB	F(2,3)	140,416	0.4667

4. Microbenchmark Detailed Perf & Statistical Validation

Measurements compiled over 30 runs per configuration comparing the Cache-Aware Fused Winograd kernel to the non-fused baseline implementation.

C_in	C_out	Fused	Threads	Mean Lat (ms)	Improv %	p-value
16	32	No	1	0.5587	-	N/A
16	32	Yes	1	0.6300	-12.77%	< 1e-10
16	32	No	4	0.5541	+0.83%	0.00012708484143657386
16	32	Yes	4	0.6298	-12.73%	< 1e-10
32	16	No	1	0.8343	-	N/A
32	16	Yes	1	0.9956	-19.33%	< 1e-10
32	16	No	4	0.8320	+0.29%	0.06921017059631815
32	16	Yes	4	0.9900	-18.66%	< 1e-10
32	32	No	1	2.5315	-	N/A
32	32	Yes	1	2.3396	+7.58%	< 1e-10
32	32	No	4	2.5383	-0.27%	0.4636329671442905
32	32	Yes	4	2.3783	+6.05%	< 1e-10
32	64	No	1	7.6729	-	N/A
32	64	Yes	1	6.1602	+19.71%	< 1e-10
32	64	No	4	7.6527	+0.26%	0.10577731874760402
32	64	Yes	4	6.1468	+19.89%	< 1e-10
64	32	No	1	3.7595	-	N/A
64	32	Yes	1	3.1474	+16.28%	< 1e-10
64	32	No	4	3.7665	-0.19%	0.23109333337777843
64	32	Yes	4	3.1487	+16.25%	< 1e-10
64	64	No	1	12.3424	-	N/A
64	64	Yes	1	8.7733	+28.92%	< 1e-10
64	64	No	4	12.3103	+0.26%	0.03914590939858788
64	64	Yes	4	8.8056	+28.66%	< 1e-10
128	128	No	1	79.0075	-	N/A
128	128	Yes	1	46.1133	+41.63%	< 1e-10
128	128	No	4	77.9719	+1.31%	< 1e-10
128	128	Yes	4	46.0179	+41.76%	< 1e-10

5. System-Level Backend Comparison

Backend	Regime (C_in x C_out)	Avg Latency (ms)	Std Dev (ms)	Speedup / Note
project	64 → 64	0.1807	0.0150	Fused Winograd F(2,3) tile microbenchmark. Overhead-dominated regime; Winograd transform cost dominates on 4x4 input.
onnxruntime	64 → 64	0.0164	0.0010	Single Conv ONNX Runtime CPU benchmark. Direct convolution is optimal on micro-inputs.
project	128 → 128	0.2736	0.0644	Fused Winograd F(2,3) tile microbenchmark. Arithmetic-dominated regime; fused path dominates on large tiles.
onnxruntime	128 → 128	5.9723	0.3014	Single Conv ONNX Runtime CPU benchmark.

6. Performance Visualization Plots

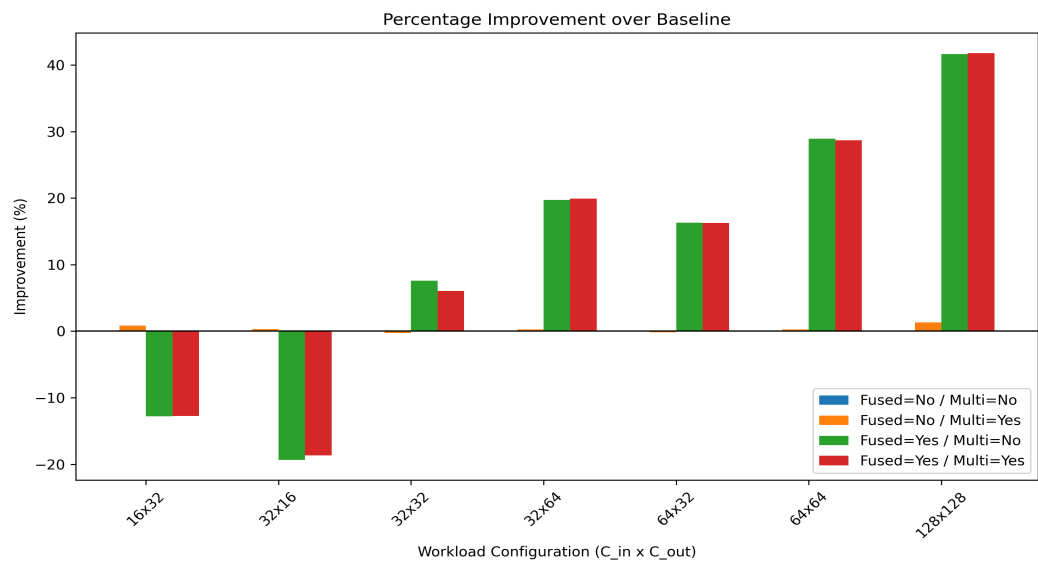


Figure 1: Percentage Latency Improvement/Regression relative to non-fused baseline across different workload shapes

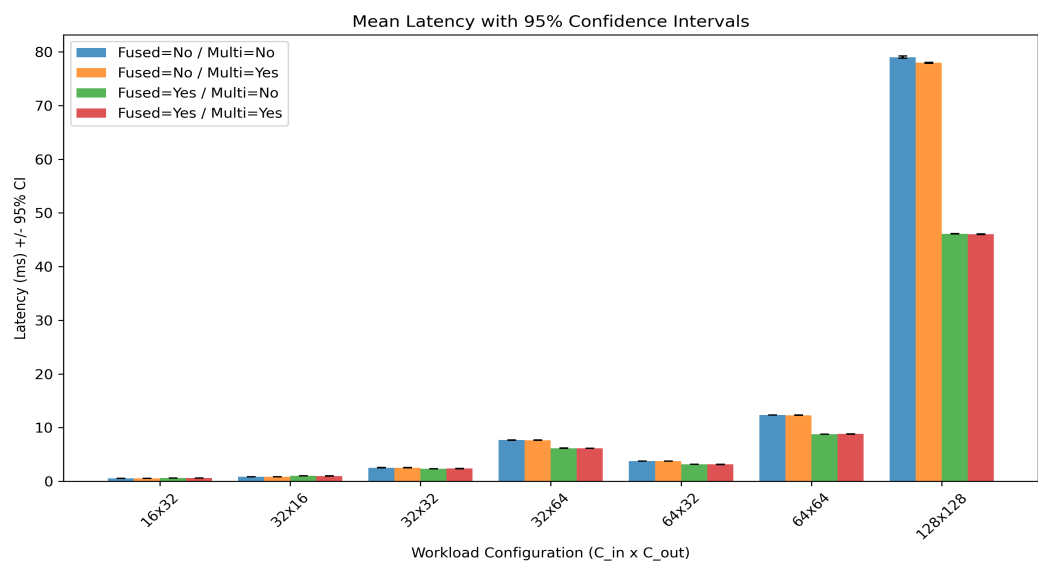


Figure 2: Confidence intervals (95%) and variance distribution across baseline and cache-aware fused runs

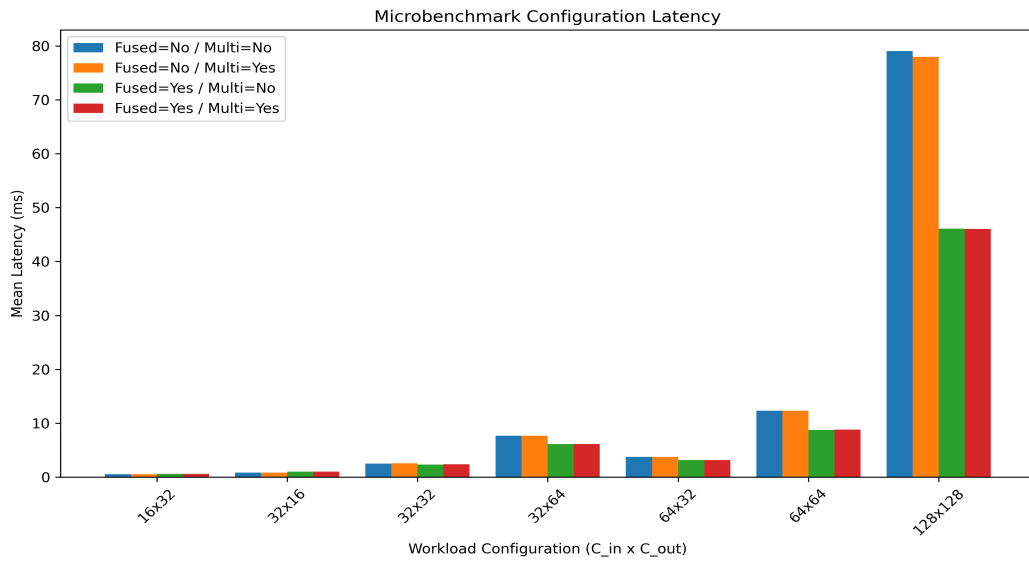


Figure 3: Absolute Latency (ms) comparison for Baseline vs Fused Winograd kernels

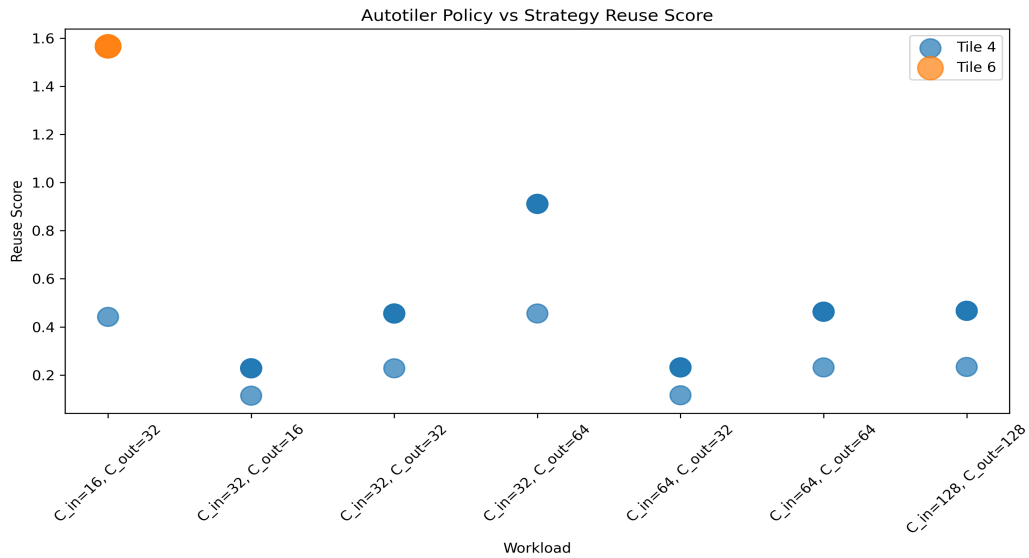


Figure 4: Autotiler working set sizes and reuse score estimates vs L1/L2 Cache Capacities

Appendix: Combined Raw Latency Run Logs

Listing of all raw latencies (ms) recorded across individual execution trials. The full dataset is printed below for complete auditability.

Config	Run	Latency	Config	Run	Latency	Config	Run	Latency
16→32 (F=F, T=1)	0	0.5677	32→32 (F=F, T=4)	10	2.5288	64→32 (F=T, T=1)	20	3.1557
16→32 (F=F, T=1)	1	0.5635	32→32 (F=F, T=4)	11	2.5460	64→32 (F=T, T=1)	21	3.1447
16→32 (F=F, T=1)	2	0.5612	32→32 (F=F, T=4)	12	2.5303	64→32 (F=T, T=1)	22	3.1481
16→32 (F=F, T=1)	3	0.5692	32→32 (F=F, T=4)	13	2.5193	64→32 (F=T, T=1)	23	3.1515
16→32 (F=F, T=1)	4	0.5546	32→32 (F=F, T=4)	14	2.5293	64→32 (F=T, T=1)	24	3.1577
16→32 (F=F, T=1)	5	0.5567	32→32 (F=F, T=4)	15	2.5150	64→32 (F=T, T=1)	25	3.1383
16→32 (F=F, T=1)	6	0.5571	32→32 (F=F, T=4)	16	2.5219	64→32 (F=T, T=1)	26	3.1423
16→32 (F=F, T=1)	7	0.5565	32→32 (F=F, T=4)	17	2.5288	64→32 (F=T, T=1)	27	3.1480
16→32 (F=F, T=1)	8	0.5558	32→32 (F=F, T=4)	18	2.5203	64→32 (F=T, T=1)	28	3.1328
16→32 (F=F, T=1)	9	0.5579	32→32 (F=F, T=4)	19	2.5240	64→32 (F=T, T=1)	29	3.1325
16→32 (F=F, T=1)	10	0.5571	32→32 (F=F, T=4)	20	2.5357	64→32 (F=T, T=4)	0	3.1478
16→32 (F=F, T=1)	11	0.5557	32→32 (F=F, T=4)	21	2.5335	64→32 (F=T, T=4)	1	3.1370
16→32 (F=F, T=1)	12	0.5660	32→32 (F=F, T=4)	22	2.5366	64→32 (F=T, T=4)	2	3.1380
16→32 (F=F, T=1)	13	0.5550	32→32 (F=F, T=4)	23	2.5355	64→32 (F=T, T=4)	3	3.1443
16→32 (F=F, T=1)	14	0.5564	32→32 (F=F, T=4)	24	2.5414	64→32 (F=T, T=4)	4	3.1475
16→32 (F=F, T=1)	15	0.5572	32→32 (F=F, T=4)	25	2.5329	64→32 (F=T, T=4)	5	3.1535
16→32 (F=F, T=1)	16	0.5577	32→32 (F=F, T=4)	26	2.5338	64→32 (F=T, T=4)	6	3.1464
16→32 (F=F, T=1)	17	0.5590	32→32 (F=F, T=4)	27	2.5328	64→32 (F=T, T=4)	7	3.1341
16→32 (F=F, T=1)	18	0.5577	32→32 (F=F, T=4)	28	2.5330	64→32 (F=T, T=4)	8	3.1360
16→32 (F=F, T=1)	19	0.5540	32→32 (F=F, T=4)	29	2.5266	64→32 (F=T, T=4)	9	3.1410
16→32 (F=F, T=1)	20	0.5646	32→32 (F=T, T=1)	0	2.3530	64→32 (F=T, T=4)	10	3.1371
16→32 (F=F, T=1)	21	0.5532	32→32 (F=T, T=1)	1	2.3462	64→32 (F=T, T=4)	11	3.1355
16→32 (F=F, T=1)	22	0.5636	32→32 (F=T, T=1)	2	2.3488	64→32 (F=T, T=4)	12	3.1332
16→32 (F=F, T=1)	23	0.5515	32→32 (F=T, T=1)	3	2.3485	64→32 (F=T, T=4)	13	3.1415
16→32 (F=F, T=1)	24	0.5549	32→32 (F=T, T=1)	4	2.3627	64→32 (F=T, T=4)	14	3.1388
16→32 (F=F, T=1)	25	0.5666	32→32 (F=T, T=1)	5	2.3401	64→32 (F=T, T=4)	15	3.1410
16→32 (F=F, T=1)	26	0.5587	32→32 (F=T, T=1)	6	2.3405	64→32 (F=T, T=4)	16	3.1467
16→32 (F=F, T=1)	27	0.5665	32→32 (F=T, T=1)	7	2.3432	64→32 (F=T, T=4)	17	3.1330
16→32 (F=F, T=1)	28	0.5560	32→32 (F=T, T=1)	8	2.3404	64→32 (F=T, T=4)	18	3.1349
16→32 (F=F, T=1)	29	0.5488	32→32 (F=T, T=1)	9	2.3436	64→32 (F=T, T=4)	19	3.1376
16→32 (F=F, T=4)	0	0.5504	32→32 (F=T, T=1)	10	2.3373	64→32 (F=T, T=4)	20	3.1303
16→32 (F=F, T=4)	1	0.5538	32→32 (F=T, T=1)	11	2.3388	64→32 (F=T, T=4)	21	3.1305
16→32 (F=F, T=4)	2	0.5485	32→32 (F=T, T=1)	12	2.3362	64→32 (F=T, T=4)	22	3.1281
16→32 (F=F, T=4)	3	0.5542	32→32 (F=T, T=1)	13	2.3434	64→32 (F=T, T=4)	23	3.1225
16→32 (F=F, T=4)	4	0.5508	32→32 (F=T, T=1)	14	2.3326	64→32 (F=T, T=4)	24	3.1133
16→32 (F=F, T=4)	5	0.5537	32→32 (F=T, T=1)	15	2.3301	64→32 (F=T, T=4)	25	3.4675
16→32 (F=F, T=4)	6	0.5507	32→32 (F=T, T=1)	16	2.3360	64→32 (F=T, T=4)	26	3.1368
16→32 (F=F, T=4)	7	0.5569	32→32 (F=T, T=1)	17	2.3430	64→32 (F=T, T=4)	27	3.1409
16→32 (F=F, T=4)	8	0.5603	32→32 (F=T, T=1)	18	2.3467	64→32 (F=T, T=4)	28	3.1628
16→32 (F=F, T=4)	9	0.5519	32→32 (F=T, T=1)	19	2.3383	64→32 (F=T, T=4)	29	3.1237
16→32 (F=F, T=4)	10	0.5534	32→32 (F=T, T=1)	20	2.3337	64→64 (F=F, T=1)	0	12.3311
16→32 (F=F, T=4)	11	0.5528	32→32 (F=T, T=1)	21	2.3317	64→64 (F=F, T=1)	1	12.3426
16→32 (F=F, T=4)	12	0.5573	32→32 (F=T, T=1)	22	2.3393	64→64 (F=F, T=1)	2	12.3474
16→32 (F=F, T=4)	13	0.5632	32→32 (F=T, T=1)	23	2.3323	64→64 (F=F, T=1)	3	12.3439

16→32 (F=F, T=4)	14	0.5557	32→32 (F=T, T=1)	24	2.3337	64→64 (F=F, T=1)	4	12.3476
16→32 (F=F, T=4)	15	0.5573	32→32 (F=T, T=1)	25	2.3342	64→64 (F=F, T=1)	5	12.3645
16→32 (F=F, T=4)	16	0.5570	32→32 (F=T, T=1)	26	2.3377	64→64 (F=F, T=1)	6	12.3251
16→32 (F=F, T=4)	17	0.5535	32→32 (F=T, T=1)	27	2.3358	64→64 (F=F, T=1)	7	12.3137
16→32 (F=F, T=4)	18	0.5517	32→32 (F=T, T=1)	28	2.3300	64→64 (F=F, T=1)	8	12.3036
16→32 (F=F, T=4)	19	0.5517	32→32 (F=T, T=1)	29	2.3299	64→64 (F=F, T=1)	9	12.3103
16→32 (F=F, T=4)	20	0.5516	32→32 (F=T, T=4)	0	2.3350	64→64 (F=F, T=1)	10	12.2717
16→32 (F=F, T=4)	21	0.5502	32→32 (F=T, T=4)	1	2.3336	64→64 (F=F, T=1)	11	12.2878
16→32 (F=F, T=4)	22	0.5552	32→32 (F=T, T=4)	2	2.3286	64→64 (F=F, T=1)	12	12.3931
16→32 (F=F, T=4)	23	0.5524	32→32 (F=T, T=4)	3	2.6444	64→64 (F=F, T=1)	13	12.3521
16→32 (F=F, T=4)	24	0.5527	32→32 (F=T, T=4)	4	2.6350	64→64 (F=F, T=1)	14	12.3600
16→32 (F=F, T=4)	25	0.5512	32→32 (F=T, T=4)	5	2.3555	64→64 (F=F, T=1)	15	12.3188
16→32 (F=F, T=4)	26	0.5606	32→32 (F=T, T=4)	6	2.4865	64→64 (F=F, T=1)	16	12.3088
16→32 (F=F, T=4)	27	0.5543	32→32 (F=T, T=4)	7	2.3783	64→64 (F=F, T=1)	17	12.3258
16→32 (F=F, T=4)	28	0.5543	32→32 (F=T, T=4)	8	2.3534	64→64 (F=F, T=1)	18	12.3230
16→32 (F=F, T=4)	29	0.5543	32→32 (F=T, T=4)	9	2.3558	64→64 (F=F, T=1)	19	12.3281
16→32 (F=T, T=1)	0	0.6338	32→32 (F=T, T=4)	10	2.3537	64→64 (F=F, T=1)	20	12.3628
16→32 (F=T, T=1)	1	0.6291	32→32 (F=T, T=4)	11	2.3546	64→64 (F=F, T=1)	21	12.3805
16→32 (F=T, T=1)	2	0.6345	32→32 (F=T, T=4)	12	2.3544	64→64 (F=F, T=1)	22	12.3704
16→32 (F=T, T=1)	3	0.6257	32→32 (F=T, T=4)	13	2.3530	64→64 (F=F, T=1)	23	12.3525
16→32 (F=T, T=1)	4	0.6271	32→32 (F=T, T=4)	14	2.3745	64→64 (F=F, T=1)	24	12.3489
16→32 (F=T, T=1)	5	0.6315	32→32 (F=T, T=4)	15	2.3748	64→64 (F=F, T=1)	25	12.3781
16→32 (F=T, T=1)	6	0.6261	32→32 (F=T, T=4)	16	2.3515	64→64 (F=F, T=1)	26	12.3960
16→32 (F=T, T=1)	7	0.6253	32→32 (F=T, T=4)	17	2.3494	64→64 (F=F, T=1)	27	12.3572
16→32 (F=T, T=1)	8	0.6302	32→32 (F=T, T=4)	18	2.3538	64→64 (F=F, T=1)	28	12.3861
16→32 (F=T, T=1)	9	0.6310	32→32 (F=T, T=4)	19	2.3556	64→64 (F=F, T=1)	29	12.3415
16→32 (F=T, T=1)	10	0.6273	32→32 (F=T, T=4)	20	2.3597	64→64 (F=F, T=4)	0	12.3015
16→32 (F=T, T=1)	11	0.6309	32→32 (F=T, T=4)	21	2.3617	64→64 (F=F, T=4)	1	12.2998
16→32 (F=T, T=1)	12	0.6271	32→32 (F=T, T=4)	22	2.3557	64→64 (F=F, T=4)	2	12.3258
16→32 (F=T, T=1)	13	0.6276	32→32 (F=T, T=4)	23	2.3752	64→64 (F=F, T=4)	3	12.3471
16→32 (F=T, T=1)	14	0.6310	32→32 (F=T, T=4)	24	2.3575	64→64 (F=F, T=4)	4	12.5116
16→32 (F=T, T=1)	15	0.6259	32→32 (F=T, T=4)	25	2.3502	64→64 (F=F, T=4)	5	12.2957
16→32 (F=T, T=1)	16	0.6262	32→32 (F=T, T=4)	26	2.3509	64→64 (F=F, T=4)	6	12.3498
16→32 (F=T, T=1)	17	0.6347	32→32 (F=T, T=4)	27	2.3532	64→64 (F=F, T=4)	7	12.3090
16→32 (F=T, T=1)	18	0.6355	32→32 (F=T, T=4)	28	2.3509	64→64 (F=F, T=4)	8	12.2914
16→32 (F=T, T=1)	19	0.6282	32→32 (F=T, T=4)	29	2.3528	64→64 (F=F, T=4)	9	12.2827
16→32 (F=T, T=1)	20	0.6270	32→64 (F=F, T=1)	0	7.5921	64→64 (F=F, T=4)	10	12.2913
16→32 (F=T, T=1)	21	0.6290	32→64 (F=F, T=1)	1	7.6477	64→64 (F=F, T=4)	11	12.2837
16→32 (F=T, T=1)	22	0.6296	32→64 (F=F, T=1)	2	7.6420	64→64 (F=F, T=4)	12	12.2786
16→32 (F=T, T=1)	23	0.6276	32→64 (F=F, T=1)	3	7.6196	64→64 (F=F, T=4)	13	12.2896
16→32 (F=T, T=1)	24	0.6295	32→64 (F=F, T=1)	4	7.6035	64→64 (F=F, T=4)	14	12.2657
16→32 (F=T, T=1)	25	0.6307	32→64 (F=F, T=1)	5	7.6139	64→64 (F=F, T=4)	15	12.2468
16→32 (F=T, T=1)	26	0.6373	32→64 (F=F, T=1)	6	7.5943	64→64 (F=F, T=4)	16	12.2381
16→32 (F=T, T=1)	27	0.6293	32→64 (F=F, T=1)	7	7.7777	64→64 (F=F, T=4)	17	12.2385
16→32 (F=T, T=1)	28	0.6337	32→64 (F=F, T=1)	8	7.6614	64→64 (F=F, T=4)	18	12.2682
16→32 (F=T, T=1)	29	0.6385	32→64 (F=F, T=1)	9	7.6497	64→64 (F=F, T=4)	19	12.2615
16→32 (F=T, T=4)	0	0.6292	32→64 (F=F, T=1)	10	7.7055	64→64 (F=F, T=4)	20	12.2484
16→32 (F=T, T=4)	1	0.6300	32→64 (F=F, T=1)	11	7.7035	64→64 (F=F, T=4)	21	12.3049
16→32 (F=T, T=4)	2	0.6379	32→64 (F=F, T=1)	12	7.6652	64→64 (F=F, T=4)	22	12.2981
16→32 (F=T, T=4)	3	0.6385	32→64 (F=F, T=1)	13	7.6786	64→64 (F=F, T=4)	23	12.3182

16→32 (F=T, T=4)	4	0.6273	32→64 (F=F, T=1)	14	7.6758	64→64 (F=F, T=4)	24	12.3102
16→32 (F=T, T=4)	5	0.6358	32→64 (F=F, T=1)	15	7.6718	64→64 (F=F, T=4)	25	12.3241
16→32 (F=T, T=4)	6	0.6270	32→64 (F=F, T=1)	16	7.6733	64→64 (F=F, T=4)	26	12.3099
16→32 (F=T, T=4)	7	0.6260	32→64 (F=F, T=1)	17	7.6672	64→64 (F=F, T=4)	27	12.3011
16→32 (F=T, T=4)	8	0.6270	32→64 (F=F, T=1)	18	7.6928	64→64 (F=F, T=4)	28	12.2950
16→32 (F=T, T=4)	9	0.6322	32→64 (F=F, T=1)	19	7.6836	64→64 (F=F, T=4)	29	12.6223
16→32 (F=T, T=4)	10	0.6288	32→64 (F=F, T=1)	20	7.6853	64→64 (F=T, T=1)	0	8.7725
16→32 (F=T, T=4)	11	0.6280	32→64 (F=F, T=1)	21	7.7053	64→64 (F=T, T=1)	1	8.7368
16→32 (F=T, T=4)	12	0.6307	32→64 (F=F, T=1)	22	7.7063	64→64 (F=T, T=1)	2	8.7302
16→32 (F=T, T=4)	13	0.6304	32→64 (F=F, T=1)	23	7.6981	64→64 (F=T, T=1)	3	8.7363
16→32 (F=T, T=4)	14	0.6272	32→64 (F=F, T=1)	24	7.7058	64→64 (F=T, T=1)	4	8.7565
16→32 (F=T, T=4)	15	0.6278	32→64 (F=F, T=1)	25	7.7075	64→64 (F=T, T=1)	5	8.7523
16→32 (F=T, T=4)	16	0.6332	32→64 (F=F, T=1)	26	7.6805	64→64 (F=T, T=1)	6	8.7358
16→32 (F=T, T=4)	17	0.6259	32→64 (F=F, T=1)	27	7.7174	64→64 (F=T, T=1)	7	8.7429
16→32 (F=T, T=4)	18	0.6345	32→64 (F=F, T=1)	28	7.6875	64→64 (F=T, T=1)	8	8.7604
16→32 (F=T, T=4)	19	0.6277	32→64 (F=F, T=1)	29	7.6738	64→64 (F=T, T=1)	9	8.7578
16→32 (F=T, T=4)	20	0.6257	32→64 (F=F, T=4)	0	7.7040	64→64 (F=T, T=1)	10	8.7428
16→32 (F=T, T=4)	21	0.6376	32→64 (F=F, T=4)	1	7.7015	64→64 (F=T, T=1)	11	8.7321
16→32 (F=T, T=4)	22	0.6280	32→64 (F=F, T=4)	2	7.6881	64→64 (F=T, T=1)	12	8.7556
16→32 (F=T, T=4)	23	0.6255	32→64 (F=F, T=4)	3	7.6675	64→64 (F=T, T=1)	13	8.7485
16→32 (F=T, T=4)	24	0.6337	32→64 (F=F, T=4)	4	7.6703	64→64 (F=T, T=1)	14	8.7330
16→32 (F=T, T=4)	25	0.6306	32→64 (F=F, T=4)	5	7.6738	64→64 (F=T, T=1)	15	8.7430
16→32 (F=T, T=4)	26	0.6240	32→64 (F=F, T=4)	6	7.6859	64→64 (F=T, T=1)	16	8.7410
16→32 (F=T, T=4)	27	0.6275	32→64 (F=F, T=4)	7	7.6792	64→64 (F=T, T=1)	17	8.7297
16→32 (F=T, T=4)	28	0.6283	32→64 (F=F, T=4)	8	7.7090	64→64 (F=T, T=1)	18	8.7309
16→32 (F=T, T=4)	29	0.6283	32→64 (F=F, T=4)	9	7.6880	64→64 (F=T, T=1)	19	8.7296
32→16 (F=F, T=1)	0	0.8396	32→64 (F=F, T=4)	10	7.6082	64→64 (F=T, T=1)	20	8.7597
32→16 (F=F, T=1)	1	0.8383	32→64 (F=F, T=4)	11	7.5843	64→64 (F=T, T=1)	21	8.7458
32→16 (F=F, T=1)	2	0.8382	32→64 (F=F, T=4)	12	7.6765	64→64 (F=T, T=1)	22	8.7232
32→16 (F=F, T=1)	3	0.8538	32→64 (F=F, T=4)	13	7.6711	64→64 (F=T, T=1)	23	9.0673
32→16 (F=F, T=1)	4	0.8350	32→64 (F=F, T=4)	14	7.6908	64→64 (F=T, T=1)	24	8.8690
32→16 (F=F, T=1)	5	0.8340	32→64 (F=F, T=4)	15	7.6650	64→64 (F=T, T=1)	25	8.8401
32→16 (F=F, T=1)	6	0.8363	32→64 (F=F, T=4)	16	7.6852	64→64 (F=T, T=1)	26	8.8666
32→16 (F=F, T=1)	7	0.8330	32→64 (F=F, T=4)	17	7.6033	64→64 (F=T, T=1)	27	8.8162
32→16 (F=F, T=1)	8	0.8362	32→64 (F=F, T=4)	18	7.6145	64→64 (F=T, T=1)	28	8.8143
32→16 (F=F, T=1)	9	0.8368	32→64 (F=F, T=4)	19	7.6120	64→64 (F=T, T=1)	29	8.8292
32→16 (F=F, T=1)	10	0.8413	32→64 (F=F, T=4)	20	7.8335	64→64 (F=T, T=4)	0	8.8202
32→16 (F=F, T=1)	11	0.8352	32→64 (F=F, T=4)	21	7.6328	64→64 (F=T, T=4)	1	8.8395
32→16 (F=F, T=1)	12	0.8319	32→64 (F=F, T=4)	22	7.6237	64→64 (F=T, T=4)	2	8.8478
32→16 (F=F, T=1)	13	0.8353	32→64 (F=F, T=4)	23	7.6302	64→64 (F=T, T=4)	3	8.8520
32→16 (F=F, T=1)	14	0.8303	32→64 (F=F, T=4)	24	7.6223	64→64 (F=T, T=4)	4	8.8183
32→16 (F=F, T=1)	15	0.8306	32→64 (F=F, T=4)	25	7.6083	64→64 (F=T, T=4)	5	8.9492
32→16 (F=F, T=1)	16	0.8358	32→64 (F=F, T=4)	26	7.5839	64→64 (F=T, T=4)	6	8.8257
32→16 (F=F, T=1)	17	0.8337	32→64 (F=F, T=4)	27	7.5840	64→64 (F=T, T=4)	7	8.8138
32→16 (F=F, T=1)	18	0.8310	32→64 (F=F, T=4)	28	7.5820	64→64 (F=T, T=4)	8	8.8406
32→16 (F=F, T=1)	19	0.8437	32→64 (F=F, T=4)	29	7.6028	64→64 (F=T, T=4)	9	8.8125
32→16 (F=F, T=1)	20	0.8336	32→64 (F=T, T=1)	0	6.1591	64→64 (F=T, T=4)	10	8.8230
32→16 (F=F, T=1)	21	0.8347	32→64 (F=T, T=1)	1	6.1653	64→64 (F=T, T=4)	11	8.7941
32→16 (F=F, T=1)	22	0.8265	32→64 (F=T, T=1)	2	6.1789	64→64 (F=T, T=4)	12	8.7842
32→16 (F=F, T=1)	23	0.8316	32→64 (F=T, T=1)	3	6.1467	64→64 (F=T, T=4)	13	8.7951

32→16 (F=F, T=1)	24	0.8260	32→64 (F=T, T=1)	4	6.1425	64→64 (F=T, T=4)	14	8.7888
32→16 (F=F, T=1)	25	0.8337	32→64 (F=T, T=1)	5	6.1522	64→64 (F=T, T=4)	15	8.7852
32→16 (F=F, T=1)	26	0.8272	32→64 (F=T, T=1)	6	6.1511	64→64 (F=T, T=4)	16	8.7931
32→16 (F=F, T=1)	27	0.8337	32→64 (F=T, T=1)	7	6.1841	64→64 (F=T, T=4)	17	8.7783
32→16 (F=F, T=1)	28	0.8266	32→64 (F=T, T=1)	8	6.1410	64→64 (F=T, T=4)	18	8.7959
32→16 (F=F, T=1)	29	0.8268	32→64 (F=T, T=1)	9	6.4282	64→64 (F=T, T=4)	19	8.7824
32→16 (F=F, T=4)	0	0.8317	32→64 (F=T, T=1)	10	6.1984	64→64 (F=T, T=4)	20	8.7748
32→16 (F=F, T=4)	1	0.8272	32→64 (F=T, T=1)	11	6.1620	64→64 (F=T, T=4)	21	8.7944
32→16 (F=F, T=4)	2	0.8314	32→64 (F=T, T=1)	12	6.1475	64→64 (F=T, T=4)	22	8.7930
32→16 (F=F, T=4)	3	0.8321	32→64 (F=T, T=1)	13	6.1547	64→64 (F=T, T=4)	23	8.7829
32→16 (F=F, T=4)	4	0.8335	32→64 (F=T, T=1)	14	6.1497	64→64 (F=T, T=4)	24	8.7803
32→16 (F=F, T=4)	5	0.8274	32→64 (F=T, T=1)	15	6.1455	64→64 (F=T, T=4)	25	8.7699
32→16 (F=F, T=4)	6	0.8424	32→64 (F=T, T=1)	16	6.1505	64→64 (F=T, T=4)	26	8.7951
32→16 (F=F, T=4)	7	0.8368	32→64 (F=T, T=1)	17	6.1464	64→64 (F=T, T=4)	27	8.7840
32→16 (F=F, T=4)	8	0.8328	32→64 (F=T, T=1)	18	6.1465	64→64 (F=T, T=4)	28	8.7788
32→16 (F=F, T=4)	9	0.8318	32→64 (F=T, T=1)	19	6.1501	64→64 (F=T, T=4)	29	8.7765
32→16 (F=F, T=4)	10	0.8313	32→64 (F=T, T=1)	20	6.1285	128→128 (F=F, T=1)	0	79.2111
32→16 (F=F, T=4)	11	0.8288	32→64 (F=T, T=1)	21	6.1514	128→128 (F=F, T=1)	1	79.1250
32→16 (F=F, T=4)	12	0.8260	32→64 (F=T, T=1)	22	6.1397	128→128 (F=F, T=1)	2	79.0692
32→16 (F=F, T=4)	13	0.8287	32→64 (F=T, T=1)	23	6.1287	128→128 (F=F, T=1)	3	78.8832
32→16 (F=F, T=4)	14	0.8292	32→64 (F=T, T=1)	24	6.1453	128→128 (F=F, T=1)	4	78.9780
32→16 (F=F, T=4)	15	0.8280	32→64 (F=T, T=1)	25	6.1414	128→128 (F=F, T=1)	5	78.8616
32→16 (F=F, T=4)	16	0.8312	32→64 (F=T, T=1)	26	6.1416	128→128 (F=F, T=1)	6	79.5193
32→16 (F=F, T=4)	17	0.8284	32→64 (F=T, T=1)	27	6.1558	128→128 (F=F, T=1)	7	79.7546
32→16 (F=F, T=4)	18	0.8404	32→64 (F=T, T=1)	28	6.1317	128→128 (F=F, T=1)	8	79.2541
32→16 (F=F, T=4)	19	0.8352	32→64 (F=T, T=1)	29	6.1417	128→128 (F=F, T=1)	9	79.5349
32→16 (F=F, T=4)	20	0.8353	32→64 (F=T, T=4)	0	6.1552	128→128 (F=F, T=1)	10	79.3098
32→16 (F=F, T=4)	21	0.8338	32→64 (F=T, T=4)	1	6.1360	128→128 (F=F, T=1)	11	79.1836
32→16 (F=F, T=4)	22	0.8270	32→64 (F=T, T=4)	2	6.2383	128→128 (F=F, T=1)	12	79.2314
32→16 (F=F, T=4)	23	0.8386	32→64 (F=T, T=4)	3	6.1697	128→128 (F=F, T=1)	13	79.2052
32→16 (F=F, T=4)	24	0.8279	32→64 (F=T, T=4)	4	6.1583	128→128 (F=F, T=1)	14	79.2417
32→16 (F=F, T=4)	25	0.8365	32→64 (F=T, T=4)	5	6.1552	128→128 (F=F, T=1)	15	79.5950
32→16 (F=F, T=4)	26	0.8293	32→64 (F=T, T=4)	6	6.1454	128→128 (F=F, T=1)	16	79.4232
32→16 (F=F, T=4)	27	0.8340	32→64 (F=T, T=4)	7	6.1903	128→128 (F=F, T=1)	17	79.5635
32→16 (F=F, T=4)	28	0.8301	32→64 (F=T, T=4)	8	6.1384	128→128 (F=F, T=1)	18	79.4571
32→16 (F=F, T=4)	29	0.8323	32→64 (F=T, T=4)	9	6.1392	128→128 (F=F, T=1)	19	79.4676
32→16 (F=T, T=1)	0	1.0001	32→64 (F=T, T=4)	10	6.1495	128→128 (F=F, T=1)	20	79.3292
32→16 (F=T, T=1)	1	0.9968	32→64 (F=T, T=4)	11	6.1333	128→128 (F=F, T=1)	21	79.2487
32→16 (F=T, T=1)	2	0.9985	32→64 (F=T, T=4)	12	6.1359	128→128 (F=F, T=1)	22	79.1226
32→16 (F=T, T=1)	3	1.0001	32→64 (F=T, T=4)	13	6.1250	128→128 (F=F, T=1)	23	77.9472
32→16 (F=T, T=1)	4	0.9952	32→64 (F=T, T=4)	14	6.1307	128→128 (F=F, T=1)	24	78.0483
32→16 (F=T, T=1)	5	0.9930	32→64 (F=T, T=4)	15	6.1273	128→128 (F=F, T=1)	25	78.4650
32→16 (F=T, T=1)	6	0.9972	32→64 (F=T, T=4)	16	6.1301	128→128 (F=F, T=1)	26	78.2677
32→16 (F=T, T=1)	7	0.9965	32→64 (F=T, T=4)	17	6.1424	128→128 (F=F, T=1)	27	78.2552
32→16 (F=T, T=1)	8	0.9987	32→64 (F=T, T=4)	18	6.1366	128→128 (F=F, T=1)	28	77.8814
32→16 (F=T, T=1)	9	0.9959	32→64 (F=T, T=4)	19	6.1411	128→128 (F=F, T=1)	29	77.7908
32→16 (F=T, T=1)	10	1.0027	32→64 (F=T, T=4)	20	6.1408	128→128 (F=F, T=4)	0	78.2245
32→16 (F=T, T=1)	11	0.9955	32→64 (F=T, T=4)	21	6.1659	128→128 (F=F, T=4)	1	77.8215
32→16 (F=T, T=1)	12	0.9942	32→64 (F=T, T=4)	22	6.1405	128→128 (F=F, T=4)	2	78.2553
32→16 (F=T, T=1)	13	0.9907	32→64 (F=T, T=4)	23	6.1337	128→128 (F=F, T=4)	3	78.2793

32→16 (F=T, T=1)	14	1.0014	32→64 (F=T, T=4)	24	6.1276	128→128 (F=F, T=4)	4	78.2943
32→16 (F=T, T=1)	15	0.9931	32→64 (F=T, T=4)	25	6.1348	128→128 (F=F, T=4)	5	78.3223
32→16 (F=T, T=1)	16	0.9929	32→64 (F=T, T=4)	26	6.1351	128→128 (F=F, T=4)	6	78.1165
32→16 (F=T, T=1)	17	0.9903	32→64 (F=T, T=4)	27	6.1407	128→128 (F=F, T=4)	7	78.0131
32→16 (F=T, T=1)	18	0.9979	32→64 (F=T, T=4)	28	6.1442	128→128 (F=F, T=4)	8	77.9538
32→16 (F=T, T=1)	19	0.9948	32→64 (F=T, T=4)	29	6.1634	128→128 (F=F, T=4)	9	77.7943
32→16 (F=T, T=1)	20	0.9964	64→32 (F=F, T=1)	0	3.7343	128→128 (F=F, T=4)	10	77.9670
32→16 (F=T, T=1)	21	0.9997	64→32 (F=F, T=1)	1	3.7334	128→128 (F=F, T=4)	11	77.8186
32→16 (F=T, T=1)	22	0.9935	64→32 (F=F, T=1)	2	3.7253	128→128 (F=F, T=4)	12	77.7952
32→16 (F=T, T=1)	23	0.9918	64→32 (F=F, T=1)	3	3.7508	128→128 (F=F, T=4)	13	77.7168
32→16 (F=T, T=1)	24	0.9895	64→32 (F=F, T=1)	4	3.8006	128→128 (F=F, T=4)	14	77.6603
32→16 (F=T, T=1)	25	0.9947	64→32 (F=F, T=1)	5	3.7680	128→128 (F=F, T=4)	15	77.6950
32→16 (F=T, T=1)	26	0.9946	64→32 (F=F, T=1)	6	3.7673	128→128 (F=F, T=4)	16	77.6853
32→16 (F=T, T=1)	27	0.9924	64→32 (F=F, T=1)	7	3.7647	128→128 (F=F, T=4)	17	78.0253
32→16 (F=T, T=1)	28	0.9998	64→32 (F=F, T=1)	8	3.7881	128→128 (F=F, T=4)	18	78.0236
32→16 (F=T, T=1)	29	0.9905	64→32 (F=F, T=1)	9	3.7992	128→128 (F=F, T=4)	19	78.0953
32→16 (F=T, T=4)	0	0.9913	64→32 (F=F, T=1)	10	3.7622	128→128 (F=F, T=4)	20	78.2780
32→16 (F=T, T=4)	1	0.9892	64→32 (F=F, T=1)	11	3.7788	128→128 (F=F, T=4)	21	78.3463
32→16 (F=T, T=4)	2	0.9992	64→32 (F=F, T=1)	12	3.7710	128→128 (F=F, T=4)	22	77.9307
32→16 (F=T, T=4)	3	0.9908	64→32 (F=F, T=1)	13	3.7737	128→128 (F=F, T=4)	23	77.9327
32→16 (F=T, T=4)	4	0.9930	64→32 (F=F, T=1)	14	3.7711	128→128 (F=F, T=4)	24	77.9106
32→16 (F=T, T=4)	5	0.9917	64→32 (F=F, T=1)	15	3.7480	128→128 (F=F, T=4)	25	78.0295
32→16 (F=T, T=4)	6	0.9909	64→32 (F=F, T=1)	16	3.7629	128→128 (F=F, T=4)	26	77.7366
32→16 (F=T, T=4)	7	0.9873	64→32 (F=F, T=1)	17	3.7611	128→128 (F=F, T=4)	27	77.5190
32→16 (F=T, T=4)	8	0.9946	64→32 (F=F, T=1)	18	3.7352	128→128 (F=F, T=4)	28	77.9412
32→16 (F=T, T=4)	9	0.9873	64→32 (F=F, T=1)	19	3.7281	128→128 (F=F, T=4)	29	77.9758
32→16 (F=T, T=4)	10	0.9846	64→32 (F=F, T=1)	20	3.7211	128→128 (F=T, T=1)	0	46.1572
32→16 (F=T, T=4)	11	0.9858	64→32 (F=F, T=1)	21	3.7267	128→128 (F=T, T=1)	1	46.1617
32→16 (F=T, T=4)	12	0.9872	64→32 (F=F, T=1)	22	3.7295	128→128 (F=T, T=1)	2	46.1064
32→16 (F=T, T=4)	13	0.9858	64→32 (F=F, T=1)	23	3.7212	128→128 (F=T, T=1)	3	46.0948
32→16 (F=T, T=4)	14	0.9861	64→32 (F=F, T=1)	24	3.7724	128→128 (F=T, T=1)	4	46.0807
32→16 (F=T, T=4)	15	0.9892	64→32 (F=F, T=1)	25	3.7782	128→128 (F=T, T=1)	5	46.1120
32→16 (F=T, T=4)	16	0.9897	64→32 (F=F, T=1)	26	3.7440	128→128 (F=T, T=1)	6	46.1097
32→16 (F=T, T=4)	17	0.9887	64→32 (F=F, T=1)	27	3.7982	128→128 (F=T, T=1)	7	46.1791
32→16 (F=T, T=4)	18	0.9953	64→32 (F=F, T=1)	28	3.7844	128→128 (F=T, T=1)	8	46.1667
32→16 (F=T, T=4)	19	0.9876	64→32 (F=F, T=1)	29	3.7845	128→128 (F=T, T=1)	9	46.3822
32→16 (F=T, T=4)	20	0.9921	64→32 (F=F, T=4)	0	3.7842	128→128 (F=T, T=1)	10	46.2468
32→16 (F=T, T=4)	21	0.9848	64→32 (F=F, T=4)	1	3.7737	128→128 (F=T, T=1)	11	46.3992
32→16 (F=T, T=4)	22	0.9916	64→32 (F=F, T=4)	2	3.7985	128→128 (F=T, T=1)	12	46.1999
32→16 (F=T, T=4)	23	0.9933	64→32 (F=F, T=4)	3	3.7814	128→128 (F=T, T=1)	13	46.3237
32→16 (F=T, T=4)	24	0.9905	64→32 (F=F, T=4)	4	3.7374	128→128 (F=T, T=1)	14	46.0885
32→16 (F=T, T=4)	25	0.9887	64→32 (F=F, T=4)	5	3.7503	128→128 (F=T, T=1)	15	46.0294
32→16 (F=T, T=4)	26	0.9901	64→32 (F=F, T=4)	6	3.7612	128→128 (F=T, T=1)	16	46.0584
32→16 (F=T, T=4)	27	0.9903	64→32 (F=F, T=4)	7	3.7807	128→128 (F=T, T=1)	17	46.0168
32→16 (F=T, T=4)	28	0.9942	64→32 (F=F, T=4)	8	3.7903	128→128 (F=T, T=1)	18	46.0818
32→16 (F=T, T=4)	29	0.9895	64→32 (F=F, T=4)	9	3.7827	128→128 (F=T, T=1)	19	46.0271
32→32 (F=F, T=1)	0	2.5048	64→32 (F=F, T=4)	10	3.7953	128→128 (F=T, T=1)	20	45.9745
32→32 (F=F, T=1)	1	2.5154	64→32 (F=F, T=4)	11	3.7923	128→128 (F=T, T=1)	21	45.9749
32→32 (F=F, T=1)	2	2.5229	64→32 (F=F, T=4)	12	3.7857	128→128 (F=T, T=1)	22	46.2616
32→32 (F=F, T=1)	3	2.5111	64→32 (F=F, T=4)	13	3.7853	128→128 (F=T, T=1)	23	46.1174

32→32 (F=F, T=1)	4	2.6453	64→32 (F=F, T=4)	14	3.7713	128→128 (F=T, T=1)	24	45.9591
32→32 (F=F, T=1)	5	2.5281	64→32 (F=F, T=4)	15	3.7731	128→128 (F=T, T=1)	25	45.9481
32→32 (F=F, T=1)	6	2.5242	64→32 (F=F, T=4)	16	3.7728	128→128 (F=T, T=1)	26	46.0924
32→32 (F=F, T=1)	7	2.5669	64→32 (F=F, T=4)	17	3.7675	128→128 (F=T, T=1)	27	45.9148
32→32 (F=F, T=1)	8	2.5003	64→32 (F=F, T=4)	18	3.7706	128→128 (F=T, T=1)	28	46.3212
32→32 (F=F, T=1)	9	2.5490	64→32 (F=F, T=4)	19	3.7697	128→128 (F=T, T=1)	29	45.8115
32→32 (F=F, T=1)	10	2.5126	64→32 (F=F, T=4)	20	3.7714	128→128 (F=T, T=4)	0	45.7097
32→32 (F=F, T=1)	11	2.5026	64→32 (F=F, T=4)	21	3.7691	128→128 (F=T, T=4)	1	45.6574
32→32 (F=F, T=1)	12	2.5145	64→32 (F=F, T=4)	22	3.7400	128→128 (F=T, T=4)	2	45.8290
32→32 (F=F, T=1)	13	2.5127	64→32 (F=F, T=4)	23	3.7353	128→128 (F=T, T=4)	3	45.8006
32→32 (F=F, T=1)	14	2.5019	64→32 (F=F, T=4)	24	3.7380	128→128 (F=T, T=4)	4	46.2620
32→32 (F=F, T=1)	15	2.5123	64→32 (F=F, T=4)	25	3.7387	128→128 (F=T, T=4)	5	46.1548
32→32 (F=F, T=1)	16	2.5588	64→32 (F=F, T=4)	26	3.7586	128→128 (F=T, T=4)	6	46.0864
32→32 (F=F, T=1)	17	2.5193	64→32 (F=F, T=4)	27	3.7438	128→128 (F=T, T=4)	7	46.1180
32→32 (F=F, T=1)	18	2.4982	64→32 (F=F, T=4)	28	3.7391	128→128 (F=T, T=4)	8	46.0722
32→32 (F=F, T=1)	19	2.5076	64→32 (F=F, T=4)	29	3.7368	128→128 (F=T, T=4)	9	45.7911
32→32 (F=F, T=1)	20	2.5045	64→32 (F=T, T=1)	0	3.1510	128→128 (F=T, T=4)	10	46.1127
32→32 (F=F, T=1)	21	2.5156	64→32 (F=T, T=1)	1	3.1622	128→128 (F=T, T=4)	11	46.1040
32→32 (F=F, T=1)	22	2.5005	64→32 (F=T, T=1)	2	3.1546	128→128 (F=T, T=4)	12	46.3668
32→32 (F=F, T=1)	23	2.5226	64→32 (F=T, T=1)	3	3.1501	128→128 (F=T, T=4)	13	45.8194
32→32 (F=F, T=1)	24	2.5467	64→32 (F=T, T=1)	4	3.1502	128→128 (F=T, T=4)	14	45.6839
32→32 (F=F, T=1)	25	2.5671	64→32 (F=T, T=1)	5	3.1490	128→128 (F=T, T=4)	15	45.7434
32→32 (F=F, T=1)	26	2.6710	64→32 (F=T, T=1)	6	3.1426	128→128 (F=T, T=4)	16	45.7943
32→32 (F=F, T=1)	27	2.5567	64→32 (F=T, T=1)	7	3.1471	128→128 (F=T, T=4)	17	46.3686
32→32 (F=F, T=1)	28	2.5284	64→32 (F=T, T=1)	8	3.1506	128→128 (F=T, T=4)	18	46.2375
32→32 (F=F, T=1)	29	2.5232	64→32 (F=T, T=1)	9	3.1758	128→128 (F=T, T=4)	19	46.1266
32→32 (F=F, T=4)	0	2.6930	64→32 (F=T, T=1)	10	3.1444	128→128 (F=T, T=4)	20	45.9945
32→32 (F=F, T=4)	1	2.5585	64→32 (F=T, T=1)	11	3.1512	128→128 (F=T, T=4)	21	46.0554
32→32 (F=F, T=4)	2	2.5205	64→32 (F=T, T=1)	12	3.1404	128→128 (F=T, T=4)	22	46.0659
32→32 (F=F, T=4)	3	2.5371	64→32 (F=T, T=1)	13	3.1431	128→128 (F=T, T=4)	23	46.2631
32→32 (F=F, T=4)	4	2.5362	64→32 (F=T, T=1)	14	3.1481	128→128 (F=T, T=4)	24	46.0445
32→32 (F=F, T=4)	5	2.5167	64→32 (F=T, T=1)	15	3.1418	128→128 (F=T, T=4)	25	46.2942
32→32 (F=F, T=4)	6	2.5271	64→32 (F=T, T=1)	16	3.1420	128→128 (F=T, T=4)	26	46.2585
32→32 (F=F, T=4)	7	2.5393	64→32 (F=T, T=1)	17	3.1410	128→128 (F=T, T=4)	27	46.0350
32→32 (F=F, T=4)	8	2.5599	64→32 (F=T, T=1)	18	3.1495	128→128 (F=T, T=4)	28	45.8545
32→32 (F=F, T=4)	9	2.5561	64→32 (F=T, T=1)	19	3.1371	128→128 (F=T, T=4)	29	45.8332